Lab 2a (Lower limb movement, EMG collection) – data collection

- **For questions about this lab please contact the TAs**
- **Files to be uploaded on Moodle:**
  1. **The data collected and inspected in this lab, together with the data processing and movement classification (done in Lab2b – data processing) will have to be uploaded to Moodle. This data will be uploaded only once. Do not upload data for Lab2a and separately for Lab2b. The report should include the names, ETH IDs, and email addresses of all students. It should be submitted as either a PDF or Word document, named "groupXX_lab2". The report should include the answers to all questions in the lab instructions (Lab2a+Lab2b).**
  2. **Python (.py) or jupyter notebook (.ipynb) files named "groupXX_lab2" with XX being the group number.**
- **Achieving a high classification accuracy is not a necessary requirement for this assignment.**

Goal

- Develop a multi-sensory wearable system to detect different movements of the lower limbs.
- Collect and process data during pre-defined movements of the lower limbs for movement classification. **Processing will be in the second part of the lab.**

Method

- Monitor leg muscle activation using EMG sensors.
- Monitor knee movements using a strain sensor.

What you will use

- Strain sensing – piezoresistive fiber sensor (fabricated during lab 1)
- Muscle activation sensing – EMG sensors (2 EMG sensors, 6 electrodes)
- Data acquisition – Arduino, voltage divider
- Breadboard, wires and 2 crocodile clips

The final wearable prototype will be made of your knee strap and 2 EMG sensors

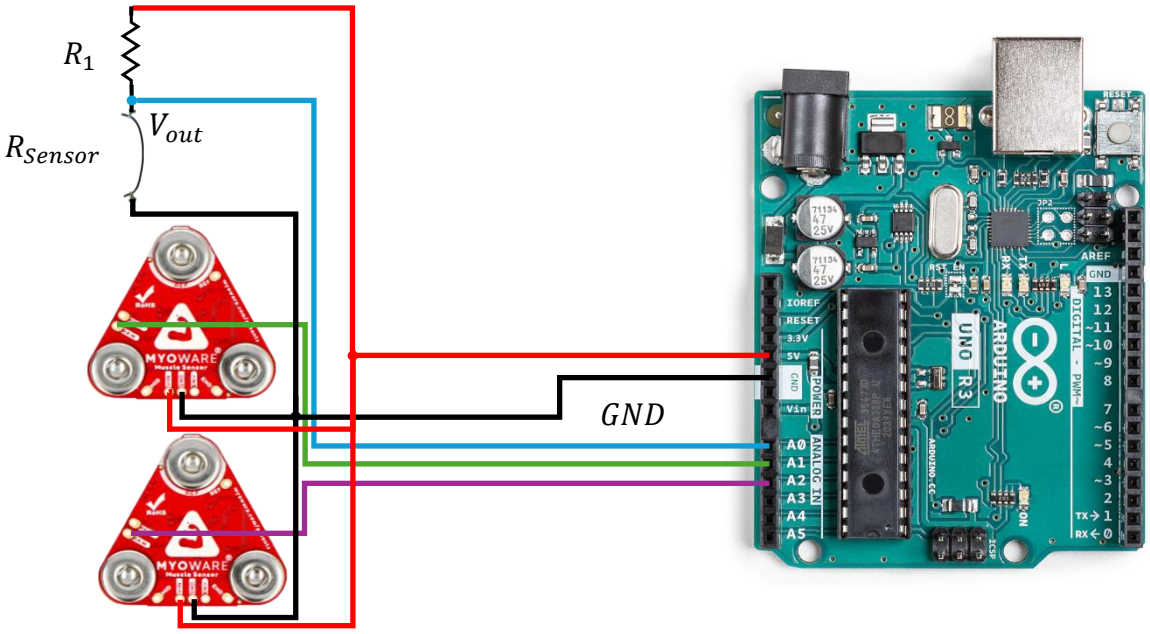
Requirements

- Laptop with a USB port with the following software installed:
  - 1. Coolterm (can be downloaded here: https://freeware.the-meiers.org/)
  - 2. Arduino IDE (can be download here: https://www.arduino.cc/en/software)
  - 3. Python with a conda environment (e.g. anaconda, miniconda)
- Each group should select the same member from lab 1 to be the participant for data collection.
  - This member should wear clothing that allows access to the skin on the lower and upper legs (shorts).

- This member will need to clean their skin with alcohol wipes in locations where medical-grade electrodes will be applied.
- This member should sign a consent form provided by us before data collection.

- Wearable device from lab 1

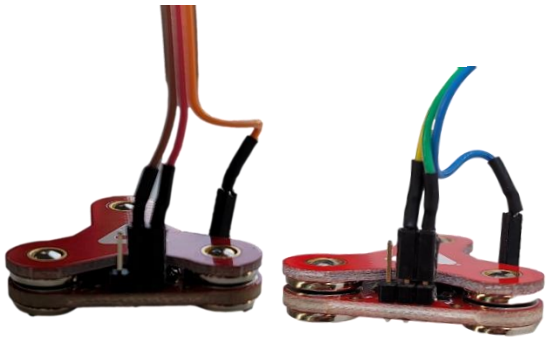
Instructions



Start with the Arduino program from lab 1 to read the strain sensor and add the following steps.

EMG data acquisition (do not attach the gel electrodes yet):

- Connect Raw pin of each EMG sensor to one of the analog input pins on the Arduino
- Connect the GND pins of the EMG sensors to the GND of the Arduino
- Connect the Vin pins of the EMG sensors to 5V on the Arduino

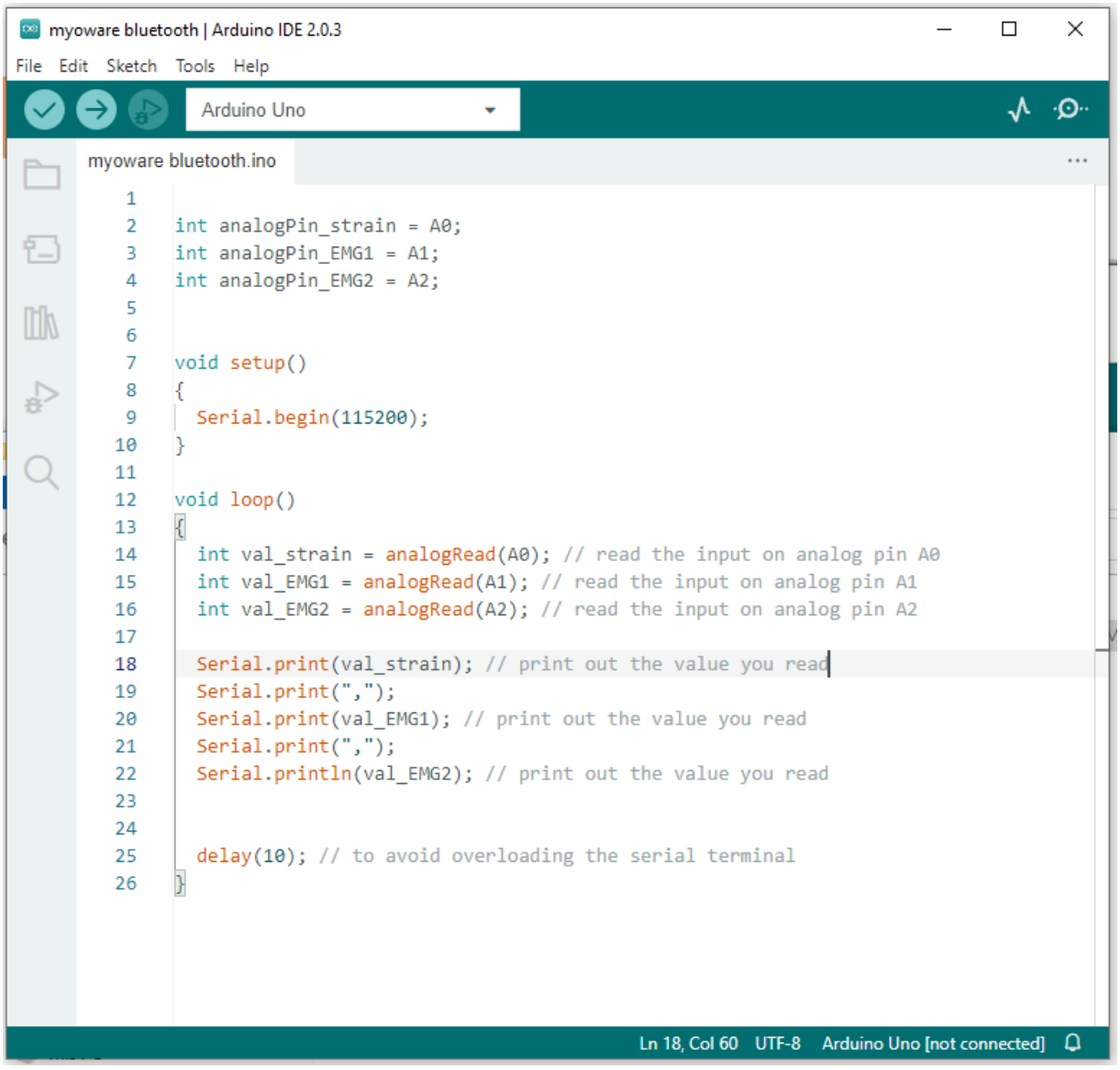


Make sure all components (including the laptop) are running on battery (not connected to wall outlet)

Arduino program:

Read the two EMG sensor outputs

- Assign the analog pins of the Arduino that are connected to EMG sensors' *RAW* pins (e.g., A1 and A2) to integers (e.g., called 'analogPin_EMG1' and 'analogPin_EMG2')
    - *int analogPin_EMG1 = A1; int analogPin_EMG2 = A2;*
- Define integers (e.g., called 'val_EMG1 and val_EMG2') to keep the values read from the EMG sensors.
    - Initialize the integers to 0. *int val_EMG1 = 0; int val_EMG2 = 0;*
- Read and print EMG output and save the value in the integers you defined in step 3.
    - In loop() add:
    - *val_EMG1 = analogRead(analogPin_EMG1); val_EMG2 = analogRead(analogPin_EMG2);*
    - *Serial.println(val_EMG1); Serial.println(val_EMG2);*



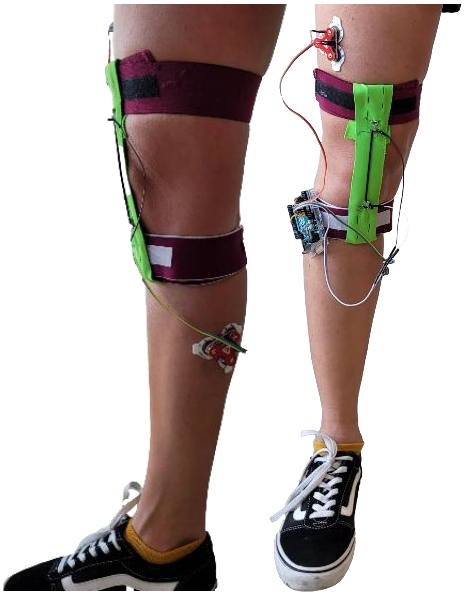
- Verify and upload the program to the Arduino
- **Note:**
- Check the values being read on serial monitor
- Close the serial monitor.

- Open the Coolterm application and connect it to your serial port
- To save data, go to Connections → Capture to Text/Binary File → Start and select the location of the file to be saved
- When your recording is done, disconnect

Sensor tests

At this point you should be able to read sensor values (both strain and EMGs) and save them to file.
Test sensor outputs: Condition the strain sensor by straining it to about 50% strain 10-20 times.

Sensor integration and application



- Use the components of the band you used in lab 1 + an extra exercise belt (with Velcro straps) to fix the strain sensor on the participant's knee as per the image below
    - Do not forget to pre-condition the strain sensor
- Connect the strain sensor and EMG sensors to your electronics based on the previous steps
- Two EMG sensors are used to monitor the activation of the quadriceps and calves muscles of one leg. Each EMG sensor is connected to three electrodes placed on the skin.
    - Clean the skin of the area where you want to apply the EMG electrodes with alcohol pad and wait for it to dry to ensure good electrical connection.
    - Apply the EMG sensors using the gel electrodes to a muscle on the participant's quadriceps and one on the calf. Refer to the image below for instructions on EMG sensor application or search https://learn.sparkfun.com/tutorials/getting-started-with-the-myoware-20-muscle-sensor-ecosystem/hardware-hookup

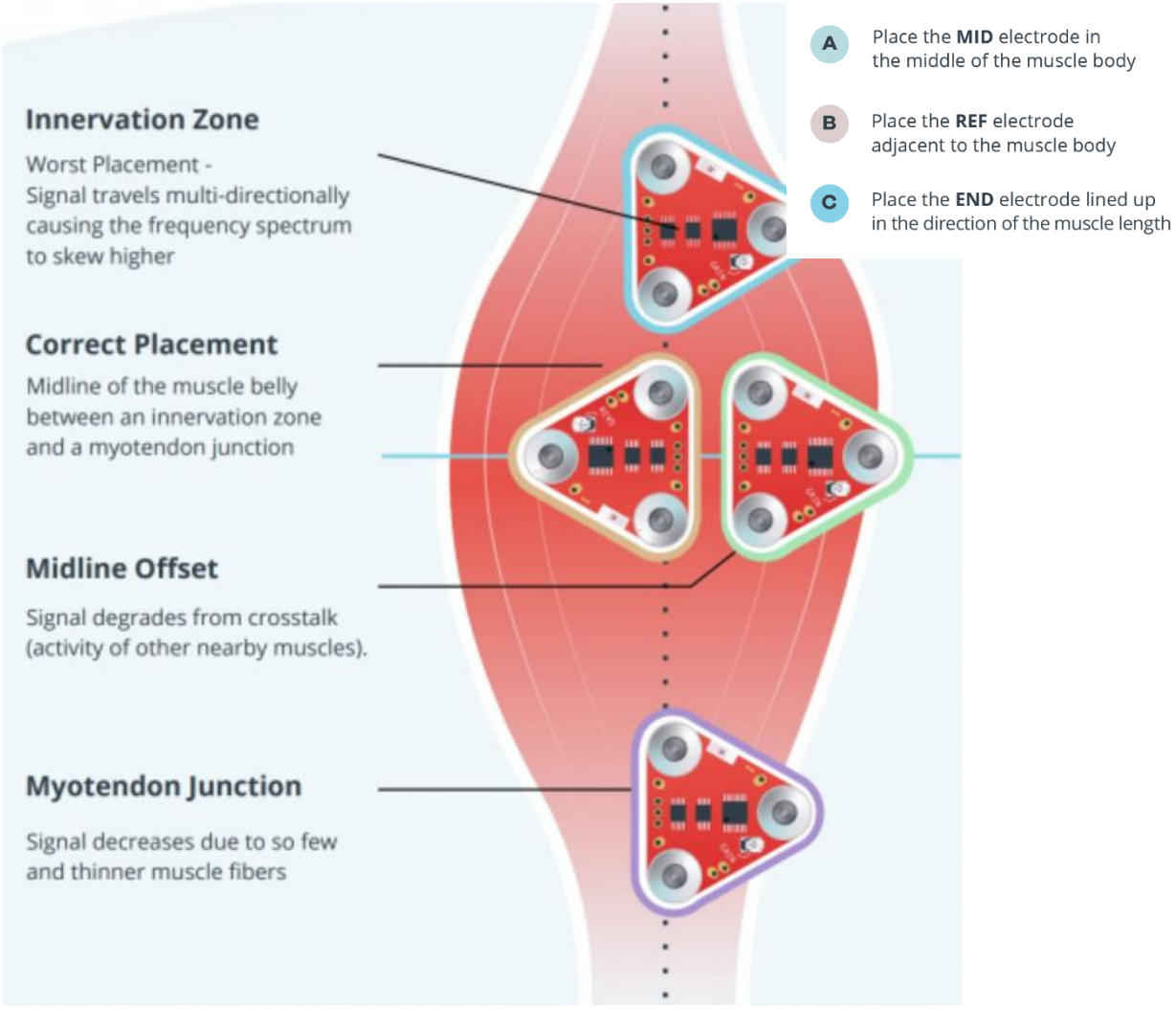


Image Courtesy of Advancer Technologies taken from the MyoWare 2.0 Advanced Guide

The innervation zone is **the region where the intramuscular terminal nerve fibers of an alpha motor neuron attach to the skeletal muscle fibers**.

The myotendinous junction is **the interface between muscle and tendon, where force is transmitted between the two tissues**.

- The green LED on EMG sensors indicate poor contact/placement. Adjust until that light is off.
- **Question 1.1** - Take a photo of your setup and upload it in your report. The positioning of the EMG sensors and strain sensors must be visible.

- In this step you will collect strain and EMG data to check that you have meaningful signals:
- Ask the participant to bend their knee with different degrees and check the placement and signal of the strain sensor.
- **Question 1.2** - What are your observations?

- Contract each muscle with the EMG electrodes and check the signals. **Question 1.3** - Plot raw EMG signals for these tests (vs. sample number) and mark the area for each test (i.e., knee bending, contraction of quadriceps, contraction of calves).
- After testing the sensors and adjusting their placements in the optimal position, data will be collected as the participant performs the following movements: stand, squat, toe raises (tip-toe stand), kick back, knee raise. For standing, just hold for 60 seconds. Each other movement should be performed 10 times. Time and take note of the time for each test for future reference.



Stand Squat Toe Raise Kick Back Knee Raise

Note that the participant should perform all movements close to the laptop which is connected to the Arduino. Be careful of connection and note that they might come off during tests.

- Save the data for each movement separately. Make sure you write down which part of collected data is for which movement. You should have the following data at the end:
    - Stand: 60 seconds
    - Squat: 10x squat
    - …
    - Knee_raise: 10x knee raise

In total you will have 5 files – one per activity.